PRODUCT SERIES

3

*Terminal Equipment*

# MUST 400

## Cardphone



INTRACOM

*HELLENIC TELECOMMUNICATIONS AND ELECTRONICS INDUSTRY*

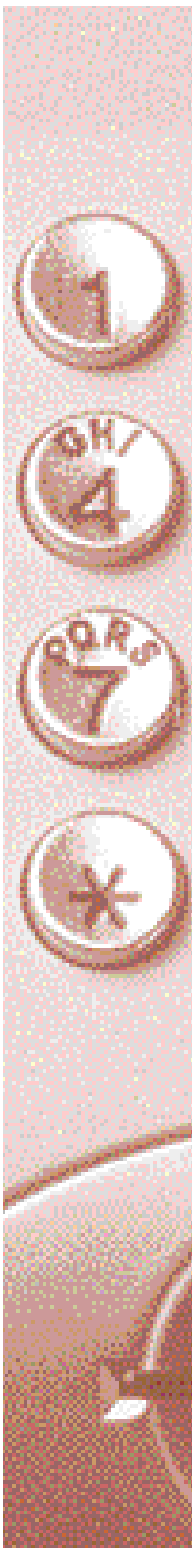
PRODUCT SERIES

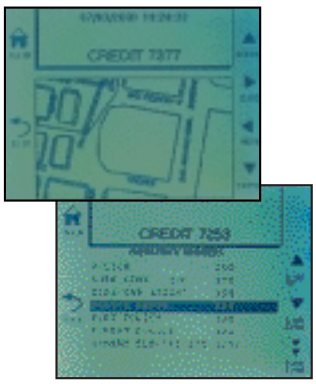
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*Terminal Equipment*

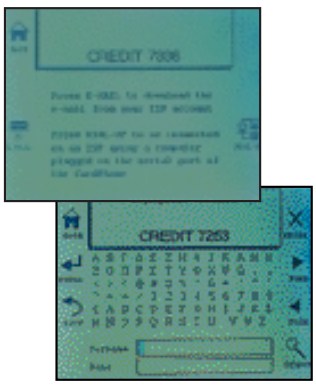
# MUST 400 Cardphone

**The MUST (Multiple Services Terminal) cardphone is the advanced solution of INTRACOM to public telephony. By utilizing the ISDN capabilities, the MUST cardphone allows the operator to provide value added services, while it retains all features of INT cardphone series. The MUST cardphone is a new member of INTRACOM cardphone system line. Supervised by the Cardphone Supervision and Maintenance System (CSMS), MUST cardphone is the ideal path to ISDN technology. For existing cardphone networks, it is a low cost solution for new ISDN installations or for direct replacement of ordinary cardphones.**









## The Terminal

MUST 400 cardphone, utilizes state of the art hardware, which is a combination of an advanced payphone plus an ISDN network terminal. Multilingual capabilities as well as all other INT payphone series services and capabilities are standard. The MUST cardphone is powered solely from the telephone line (no battery is required) and follows INTRACOM philosophy of modular structure for easy in the field maintenance. The large LCD screen (1/4 VGA) can display, in a user-friendly interface, content either in text or graphical format. The MUST uses the ISDN line (voice and data) at 2x64 Kbit data transfer rate. Having an inherent high data rate, the cardphone can communicate with the Content Formatting and Distribution System (CFDS), which manages and formats the content that will be displayed on the terminal in order to provide up-to-date information to the user.

## Content Formatting and Distribution System (CFDS)

This new generation of INTRACOM' s cardphones comes along with a new service and content management system (CFDS). CFDS is fully integrated with the existing Cardphone Management System (CSMS), extending in this way the overall capabilities of the cardphones network and protecting at the same time the initial investment of the operator. The system servers accept requests by the terminals and receive information from other content provision systems and finally send the relevant content to the terminal, after having formatted the output in the most suitable way. Security of communication is ensured using an AAA server. This server provides Authentication, Authorization and Accounting services for the MUST cardphones protecting the system against intentional or

accidental abuse and unauthorized access. In addition, the Content Provider has control over the content displayed and may create new income opportunities by leasing advertisement space on the ISDN terminal screen. CFDS architecture is a two layered structure. In the first layer, the National CFDS provides the communication with the Content Provider System and is responsible for distributing the content to the Regional CFDS servers. In the second layer, the Regional CFDS servers provide the formatting and the downloading of the content to the ISDN terminals.

## Services

CFDS is implemented in a way to provide services according to operator' s needs. Priority has been given to flexibility. More specifically, the type of content that can be displayed on the ISDN cardphone screen includes:

- Static information, which is stored in cardphone' s memory depending on the location of the terminal.
- Dynamic information, depending on user input, for which communication with a server is required.

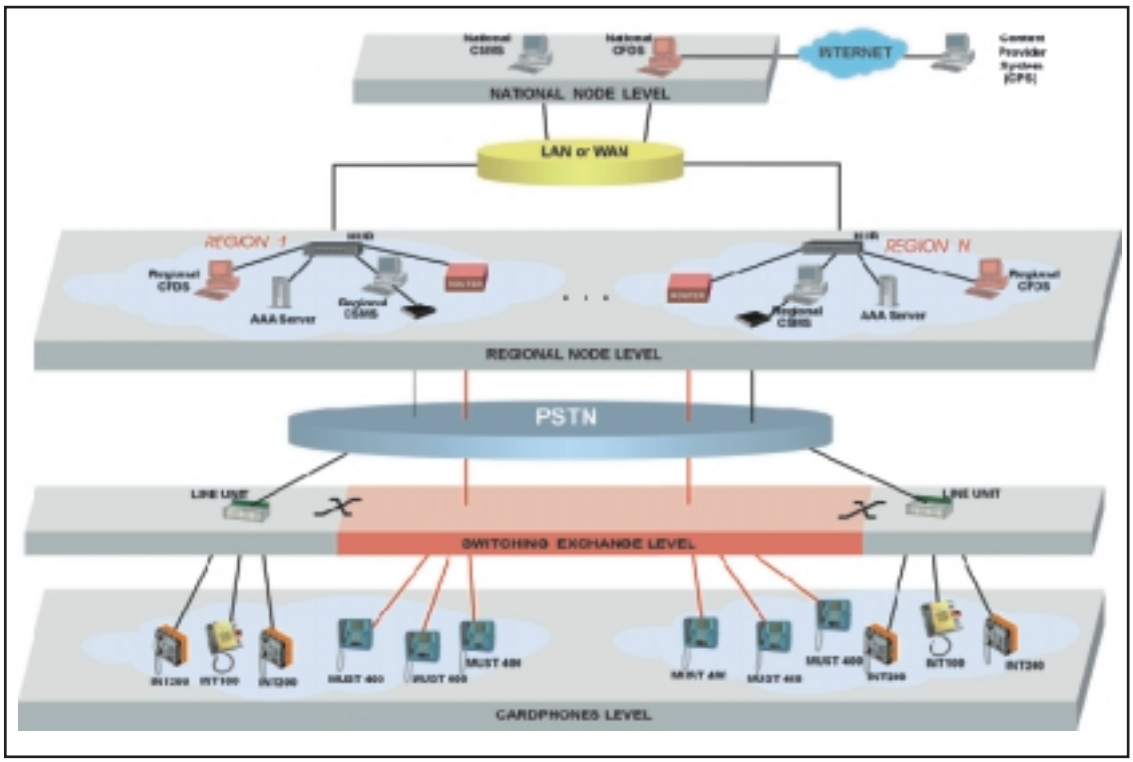
Access to all services is simple and straightforward. Eight programmable buttons, located at both sides of the MUST terminal screen, control all actions. The functionality of each button depends on the screen being displayed. Each button is associated with a descriptive image and short text, making navigation fast and easy. Access to services can be achieved during the idle mode or during the call. Examples of services provided by the MUST 400 are:

- Telephone Directory
- Weather Forecast
- SMS Sending
- E-mail reception (text) or external PC connection
- Neighborhood map
- On Duty Emergency Places (e.g. Hospitals, Pharmacies)



- Emergency Numbers (e.g. Police, Firebrigade)
- Stock Exchange Information
- Exchange rate information

The operator has the ability to create new services and the associated screens, simulate the operation and decide the group of cardphones these services will be applied to. During the on-hook state, the MUST cardphone provides the capability to display advertising messages, in graphics format, which are of global or local content.




**Cardphone System Architecture**

# Technical Specifications

## POWER:
- Line powered from the twisted copper pair line (no battery is required)

## NETWORK:
- ISDN U-interface
- In compliance with ETSI ETR080, ANSI T1 - 601-1992, FTCS3211
- Twisted copper pair line

## DEBITING:
- As per ISDN specification
- Self-debiting capability for the telephone calls
- Flexible debiting for the services
- Simultaneous debiting capability for services and calls

## SECURITY
- Eurochip Active authentication
- "White" and "Black" list of telephone cards
- Transaction monitoring

## HOUSING:
- Aluminum Cast
- Special treated against corrosion.
- Electrostatically painted with polyester type powder paint.

## KEYPAD:
- Twelve (12) standard keys for dialing according to ITU-T recommendations.
- Four function keys (software programmable) in order to perform operations such as change language, card change, follow on call, redial, volume control, etc.
- Vandal proof

## CARD READER:
- Modular card reader for optimal maintenance and replacement
- Protection against insertion or foreign objects

## PHYSICAL CHARACTERISTICS:

| | |
|---|---|
| **Dimensions:** | H 290mm x W 285mm x D 155 mm (with handset) |
| **Weight:** | 6 Kg |
| **Environmental:** | -30˚C to +65˚C<br>Up to 95% RH at 40˚C |

*Specifications subject to change without prior notice*

03.01

# Cardphone Features

## CALL TYPES
- Charged outgoing calls: local, long distance international, mobile calls
- Free of charge calls
- Barred calls
- Incoming calls

## CARD
- Prepaid telephone card, chip card technology, EUROCHIP I & II, T2G, T2G+
- Five sockets for Security Modules (SAM)
- Subscriber card
- Optionally magnetic credit card, track 2

## USER INTERFACE
- User friendly
- Large LCD display (1/4 VGA)
- Multi-language support
- Card change, follow on call, volume control, redial keys
- Hearing aid coupling

## MAINTENANCE
- Modular structure
- Parametrical operation
- Self test capability
- Supervision from remotely placed Management System (CSMS)
- Alarm calls in case of serious malfunction

## ADDITIONAL FEATURES
- Complete S/W downloading
- Service log

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